MAHATMAGANDHIUNIVERSITY,KOTTAYAM

MGU-UGP (HONOURS)

THIRDSEMESTEREXAMINATION
(2024ADMISSIONONWARDS)

MG3DSCECT204 DIGITAL FUNDAMENTALS

Duration:1.5hours MaximumMarks:50

PartA
MultipleChoiceQuestions
Answer **All** Questions.
Eachquestioncarries**1**mark

Writecompleteanswersofyourchoiceintheanswerpaperagainsteachquestionnumber

1.Whichofthe followingrepresentsthedecimalnumber45 in binary? U 1

a)101001 b)101101 c)110101 d)100101

2. The1'scomplementofthebinarynumber100110 is: U 1

a)011010 b)011001 c)111001 d)100101

3. WhichlogicgateproducesaHIGHoutputonlywhenallinputsare HIGH? U 2

a)OR b) AND c)NOR d) XOR

4. Identifythecorrectexpressionfor DeMorgan'sfirsttheorem fromtheoptionsgiven below.

a) $\overline{A+B} = \overline{A} + \overline{B}$
b) $\overline{A \cdot B} = \overline{A} + \overline{B}$
c) $\overline{A+B} = \overline{A} \cdot \overline{B}$
d) $A \cdot \overline{B} = \overline{A} + \overline{B}$

U 2

5. In a4-variableK-map,howmanycellsarepresent? U 2

a)4 b) 8 c)12 d)16

6. JKflip-flopwithJ=1,K=0will: K 3

a)Resettheoutput b)Toggle theoutput

c)Set theoutput to 1 d)Holdtheoutput

7. In aPISOshiftregister,datais: K 3

a)Parallelin,Serialout b)Parallelin,Parallelout

c)Serialin,Serialout d)Serialin,Parallelout

8. Whichmemorytypeisnon-volatileand storesdatapermanently? U 3

a)SRAM b) DRAM c)ROM d) Cache

9. Booth'smultiplicationalgorithmismainlyusedfor: U 3

a) Additionofbinarynumbers

b) Fastmultiplicationofsignedbinarynumbers

c) Divisionofbinarynumbers

d) Floatingpointaddition

10. Whichcontrolunit typeusesasequenceofmicroinstructionsstoredin memory? U 3

a) Hardwiredcontrol b)Microprogrammed control

c)Pipelinedcontrol d)Distributedcontrol

1 x 10 = 10

PartB

ShortAnswerQuestions

Answer **4** Questions

Eachquestioncarries**5**marks

11. Convertthedecimal number375into binary,octal, andhexadecimalrepresentations.U 1

12. Applythelogicgateprinciplestoconstructatruth tableforan XORgateanddemonstrateitsuse in generating odd parity. A 2

13. SimplifytheBooleanexpressionusingKarnaughMap:F(A,B,C,D)=Σm(0,2,5,7,8,10,13,15) U 2

14. Explaintheworking principleof aMaster-SlaveJKflip-flopwith aneat diagram U 3

15. Classify SRAM, DRAM, and Cache memory based on their speed, volatility, and cost, and illustratetheirusageina computer system. A 3

16. IllustratethedifferencebetweenHardwiredcontrolandMicroprogrammedcontrol. A 3

4 x 5 = 20

PartC

Essay

QuestionsAnswerany**2**Q

uestions

Eachquestioncarries**10**marks

17. Illustratetheoperationofbasiclogicgatesbyconstructingtheirtruthtables,drawingtheirsymbols, and generating pulsed waveforms. A 2

18. Construct the logic diagrams and truth tables for RS, JK, T, and D flip-flops, and explain their operationwiththehelpofsequentialcircuit concepts. A 3

19. Explaintheconstructionandworkingofdifferenttypesofshiftregisters(SISO,SIPO,PISO,and PIPO) with neat diagrams. U 3

2 x 10 = 20